JIANXUAN SHAO

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RESEARCH INTERESTS

Human-AI Interaction | Proactive AI Assistance | Context-Aware Support | Interaction Design and Evaluation

Proposed doctoral research: Investigating how AI assistants use task context and user feedback to decide whether, when and how to offer help before an explicit request, while preserving user control.

EDUCATION

University of Sydney Feb 2025 - Present

Master of Computer Science (Data Science and AI), expected Dec 2026

Average across 12 graded units: 79.33/100; Semester 1, 2026 average: 84.5/100. Selected results: Machine Learning and Data Mining 89 HD; Introduction to Programming 87 HD; Data Structures and Algorithms 77 DI; Introduction to Cybersecurity 92 HD; Computer and Network Organisation 85 HD.

Hebei University Sep 2018 - Jun 2023

Bachelor of Engineering in Architecture

GPA 4.05/5.0. Five-year degree covering architectural design, architectural physics, urban planning, computational design and digital visualisation.

SELECTED RESEARCH AND COMPUTATIONAL PROJECTS

Explainable AI-Generated Image Detection 2026 - Present

Research design and analysis lead | University of Sydney | Eight-student capstone team

- Led research framing, experimental planning and result analysis; implemented the direct detection baseline. Teammates implemented other branches, integrated fusion and conducted module-level evaluations.
- Combined two Qwen3.5-9B routes, verdict before explanation and evidence before verdict, with ResNet-50 visual forensics through logistic fusion. FakeBench evaluation used 6,000 images: 3,000 real and 3,000 fake from 10 generators.
- Team result: fusion achieved 94.33% balanced accuracy on the 900-image evaluation split, 5.67 percentage points above ResNet-50, the strongest single branch.
- Exploratory ablations refitted only the logistic fusion classifier on validation-set predictions. Removing ResNet-50 reduced evaluation balanced accuracy by 7.22 points; removing evidence-before-verdict reasoning reduced it by 0.44, with the latter difference not statistically confirmed.
- Analysed detection errors alongside generated explanations. Better text similarity did not yield statistically confirmed explanation-quality gains; explanations were assessed on 200 fake images using automatic metrics and a model judge.
- The language model was selected using scores that included this evaluation split; ablations were post hoc. External testing and human explanation evaluation remain future work.

AI Game Companion - Game Fellows Initiated 2025

Prototype completed | Prototype development and study preparation

- Developed a Python/Pygame co-play testbed by extending a Galaga-style game with a visible, rule-based teammate for threat interception, shielding and coordinated support fire.
- Linked text/audio callouts to support actions; used cooldowns and stable behaviour states to limit repeated prompts and rapid switching.
- Designed and implemented counterbalanced Solo/Fellow trials with paired random seeds and gameplay/event logs. Prepared questionnaires on presence, trust, usefulness and player control; no recruited-user study was conducted.

Machine Learning Classification and PathMNIST Pipelines 2026

COMP5318

- Built classification workflows with preprocessing, stratified cross-validation and hyperparameter tuning; extended the pipeline to PathMNIST using Random Forest, MLP and CNN baselines.
- Compared accuracy, macro-F1, confusion matrices and runtime trade-offs, with attention to class-level errors and model behaviour.

Electricity Sector Data Engineering and Spatial Integration 2026

COMP5339

• Integrated electricity, emissions, renewable-station and regional data with geocoded coordinates and spatial DuckDB storage. Processed 716,454 facility records and streamed JSON via MQTT to a Streamlit map dashboard.

BUILT ENVIRONMENT AND INTERACTIVE TECHNOLOGY EXPERIENCE

Shanghai Fengyuzhu Culture Technology Co., Ltd. Jul 2023 - Jul 2024

Interactive Media Intern

• Supported immersive spatial systems integrating CAVE projection, wall-touch interfaces, radar recognition and VR/AR visualisation, connecting architectural environments with interactive digital technologies.

• Worked across design, technical delivery and on-site system debugging, translating spatial concepts into user-facing interactive experiences.

Hebei Haoyu Architectural Design Co., Ltd. - Beidaihe Branch Jun 2022 - Aug 2022

Architectural Design Intern

• Contributed to an experimental primary-school project through site and user-needs research, spatial layout studies, client feedback iterations and design development.

• Produced floor plans, elevations, sections and visualisations in CAD-based architectural workflows.

COMPUTATIONAL DESIGN AND ARCHITECTURAL EXPERIENCE

Surveying and Digital Reconstruction of Daci Pavilion and Geyuan Temple Oct 2021

Team Member

• Recorded historic architectural geometry through sketches and CAD/Rhino documentation, reconstructing plans, elevations, sections and three-dimensional information for preservation.

• Modelled architectural components including traditional bracket forms; project records were archived in Baoding City Library.

Qinhuangdao Guangjian Construction Project Management Co., Ltd. Jun 2021 - Aug 2021

Intern / Assistant Architect

• Produced demolition, facade, floor-plan and sectional drawings for a shantytown redevelopment project, following senior architects' design requirements.

ADDITIONAL TECHNICAL AND INTERDISCIPLINARY EXPERIENCE

Mobile security analysis (INFO5995): Threat modelling and reproducible technical reporting of Android authentication logic.

Computational design training: Rhino/Grasshopper workshops, SCI-Arc digital visualisation workshop, UIA-HYP competitions (2021, 2022).

TECHNICAL SKILLS

Programming and prototyping: Python, SQL, interactive prototyping and data processing.

Research and evaluation: Reproducible model evaluation, error analysis and controlled study preparation.

Design and interaction: Computational and spatial design, 3D modelling and immersive interfaces.

AWARDS AND LEADERSHIP

Hebei University Outstanding Student Scholarship - Second Prize (2019); Third Prize (2021, 2022).

JYZC Undergraduate Innovation & Creativity Contest - Nomination Award (2022).

President, Hebei University Tennis Club (2019-2020); Team Leader, Hebei University architecture-themed campus sketching activity (2022).